Reflection - Module 10 Lab: Unsupervised Learning

Working through the Module 10 Unsupervised Learning Lab really helped solidify the shift in mindset between supervised and unsupervised machine learning. Up to this point, every model I built relied on labeled target variables (y) to guide training and measure accuracy. Stepping into unsupervised learning meant letting the algorithms discover hidden structure on their own using only the input features (X).

When applying K-Means clustering to the synthetic customer data, the biggest takeaway for me was understanding the balance between minimizing inertia and avoiding overfitting. While increasing k will always push inertia closer to zero, plotting the Elbow Method made it visually clear why k=4 was the true sweet spot where adding more clusters stopped providing meaningful separation. Similarly, running Principal Component Analysis (PCA) on the 4-dimensional Iris dataset showed me how powerful dimensionality reduction is in practice. Seeing that just two principal components captured 97.77% of the total variance (92.46% in the first and 5.31% in the second) proved that the original four features were heavily correlated and carried redundant information. Compressing the data down to 2D made it easy to plot and visually separate the species without losing the underlying patterns. Beyond the machine learning concepts, troubleshooting the single-line formatting issues and missing library imports (seaborn) in the starter notebook was a good practical reminder that debugging environment and syntax quirks is just as much a part of the workflow as building the models themselves.

AI Disclosure:
What AI was used: Google Gemini.
Why it was used: To help troubleshoot syntax errors caused by single-line formatting in the starter notebook cells and to structure the code blocks for K-Means and PCA.
What I learned in reflection: I learned how the Elbow Method balances low inertia against overfitting when selecting k, and how capturing 97.77% of variance in just two PCA components proves that the original four Iris features are highly correlated.
How I modified it to make it my own: I executed and verified each code block in my Google Colab environment, added the missing seaborn import to generate the PCA visualization, analyzed the exact variance output from my runtime, and committed the final .ipynb file to my GitHub repository.